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An observation of the food preference of freshwater bivalve, *Lamellidens marginalis* (Lamarck, 1819) of Patamda block, E. Singbhum

Amrita Dutta Nayak*
Department of Zoolgy, Ranchi University, Ranchi, Jharkhand, India

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Abstract :*Lamellidens marginalis* (L) is a commonly occurring bottom dwelling, freshwater bivalve. It feeds on a variety of food like aquatic plants, zooplankton, algae as well as mud & debris. The percentage of the different food items however depends on its availability due to seasonal factors.

Keywords : *Lamellidens marginalis* (L.), bivalve, freshwater, gut contents, zooplankton, algae, aquatic plants, mud.

INTRODUCTION

Lamellidens marginalis (Lamarck, 1819) is a common freshwater bivalve species widely distributed in the lower and upper Gangetic plains in India, Bangladesh, Sri Lanka, Myanmar etc. It is usually found in dense population in stagnant water bodies like ditches, ponds, lakes, rivers etc.

It is among the most preferred food species and is also used in medicines. The shells are used to extract edible lime.

L.marginalis is a commonly occurring bivalve in the Singbhum Plateau. It usually occurs in association with dense aquatic vegetation and algal growths. Availability of preferred food along with other physio-chemical factors decides the growth of any organisms. The distribution and abundance of *L. marginalis* species in a given habitat is affected by food availability.

The present observation of food preferences of *L.marginalis* was carried out in some freshwater lentic water habitats of Patamda block, East Singbhum, Jharkhand. The Patamda block is an amalgamation of 84 big and small villages (according to 2011 census) and is blessed with widespread natural beauty in form of freshwater ponds, ditches, pools, paddy fields, rivers, streams as well as manmade dams, lakes and reservoirs etc which hosts a good population of *L. marginalis.*

*Corresponding author :
Phone : 8210745001
E-mail : amritadutta50@gmail.com

MATERIALS & METHODS

The collection of the bivalve *L. marginalis* was carried out from March 2011 to Feb 2012, at Patamda block, E. Singbhum.

Sampling methods:

Molluscs were collected either by careful hand picking or by using hand nets and sieve nets . Live specimens were preserved at 4% formalin solution and kept in different vials bearing tags denoting collection date , time and respective sites . Some dry shells were also collected at the banks of the water bodies, which were cleaned and stored in wooden box for further study. Collections were made during the morning and evening hours.

For the analysis of food contents of the bivalve, the shell were broken to expose the animal body and alimentary canal. Latter on stomach contents were taken out by making an incision in the alimentary canal and eventually stored in 4% formalin solution.

2 methods were used to determine the composition of the consumed food items - Numerical method and volumetric method.

In the numerical method, different varieties of food items were segregated and then total count is worked out.

In the Volumetric methods, total volume of the entire stomach is determined by water displacement method. Finally, different food items are segregated and volume of each one is obtained. This represents the relative importance of each food items.

RESULTS & DISCUSSION

Table 1- Various types of identified food items in gut of *L.marginalis*

Species	Food items identified in the gut/ stomach
Lamellidens marginalis	1. Aquatic plants 2. Fine algae 3. Zooplanktons (including Protozoans, Rotifers, copepods etc) 4. Debris 5. Mud

Table 2- Relative Importance of major food items in different months in the gut/stomach of *L.marginalis*

Food items	Mar' 11	Apr' 11	May' 11	Jun' 11	Jul' 11	Aug' 11	Sep' 11	Oct' 11	Nov' 11	Dec'11	Jan' 11	Feb' 11
Detritus	7.00	7.70	9.20	18.40	19.10	14.30	_	13.25	9.00	5.00	3.90	5.20
Aquatic Plants	21.00	30.00	40.00	50.10	30.20	25.10	29.60	20.30	19.50	22.60	23.90	19.50
Algae	14.50	12.50	7.10	9.50	12.10	10.30	9.00	15.10	12.00	18.00	18.50	16.00
Zoo-plankton	20.10	19.20	18.60	16.20	41.00	47.10	50.30	48.70	30.60	25.20	30.102	28.20
Mud	30.10	22.90	20.20	18.40	9.50	10.50	15.90	12.00	15.00	23.80	25.90	26.70

It is observed that as *L.marginalis* is a bottom dweller , some amount of mud and detritus was observed among the stomach contents. However, the different food items reveal the multiple choices of items that the bivalves chooses as food . The result of Table- 2 also reveals that the major portion of food items in the monsoon months was made up by the zooplankton groups but after the monsoon especially in summer and winter, zooplankton percentage as a chief food item decreased considerably. On the other hand, simultaneously the percentage of aquatic plants increased to cope up the decreasing amount of zooplankton. This may be due to the availability of zooplankton during the monsoon season. However, soon after the flood season is over, there is a decrease in availability of zooplankton .

This is, however, coped up by having alternative food preferences like algae which is available round the year which is shown by its percentage in the food. However, alteration may be in case of different types of algae like filamentous algae etc. that is consumed as food.

Another preferable food item seems to be the detritus which comprises the leaf litter, debris etc. The important of leaf litter in food preferences of L. *marginalis* probably lies in providing an energy source of microbial growth.

Bivalves are filter feeders and degree of opening of shell valves and extensions of organs like foot for burrowing and siphons for active feeding depends upon the existing environment . This is in accordance that different food groups showed a seasonal rise or low in the gut contents.

The present observation is merely a preliminary investigation in finding out what gets into the stomach of the mussel but it does not identifies what is digested , absorbed and assimilated. The present study has analysed what the mussel is consuming. However, gut / stomach content analysis does not differentiate between what is being digested and what will pass through the mussel.

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